



Embedded Systems Course – ADC programming with PSOC

Prakash Balasubramanian
Boya Vinay Kumar



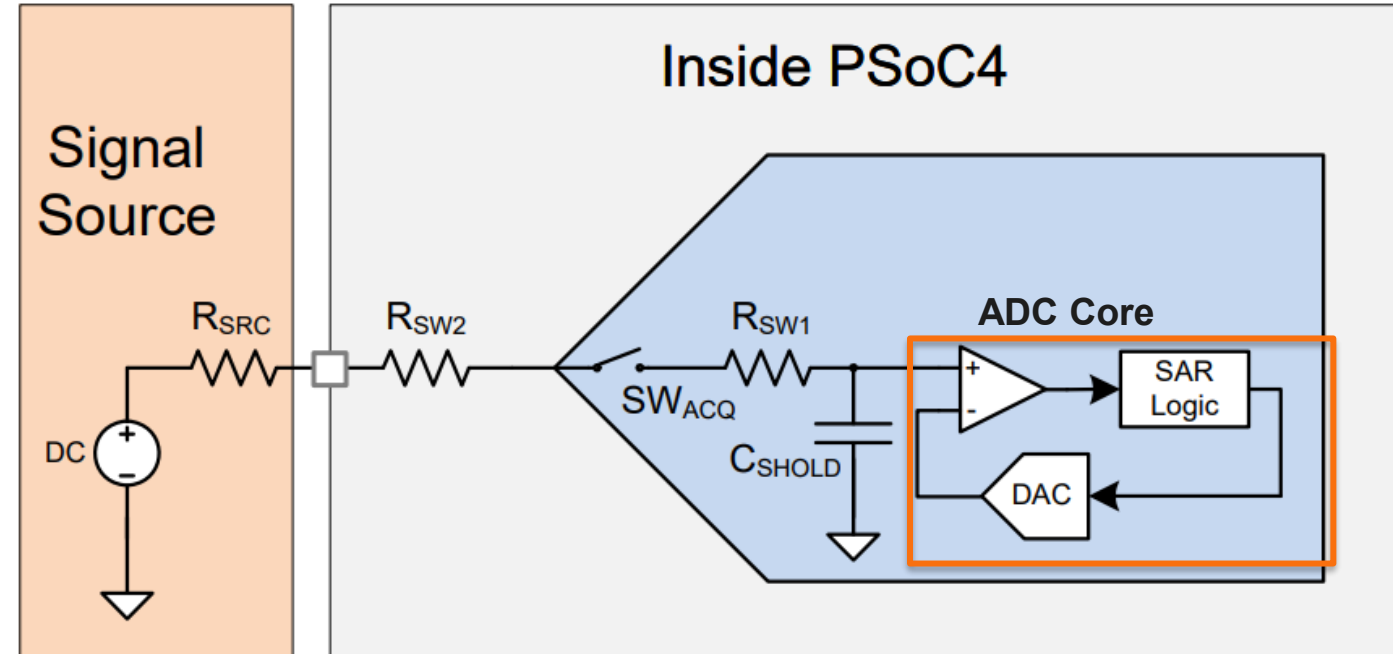
PSOC SAR ADC

- **Sample** – Sampling the input signal.
- **Hold** – the action of holding the voltage for a brief amount of time by use of a capacitor.
- The recommended value of acquisition time is:

$$t_{ACQ} \geq 9 \times (R_{SRC} + R_{SW2} + R_{SW1}) \times C_{SHOLD}$$

where: $R_{SW2} + R_{SW1} = \sim 1000 \text{ Ohms}$

$C_{SHOLD} = \sim 10 \text{ pF}$



SAR ADC Architecture

SARMUX :

SARMUX is an analog dedicated programmable multiplexer

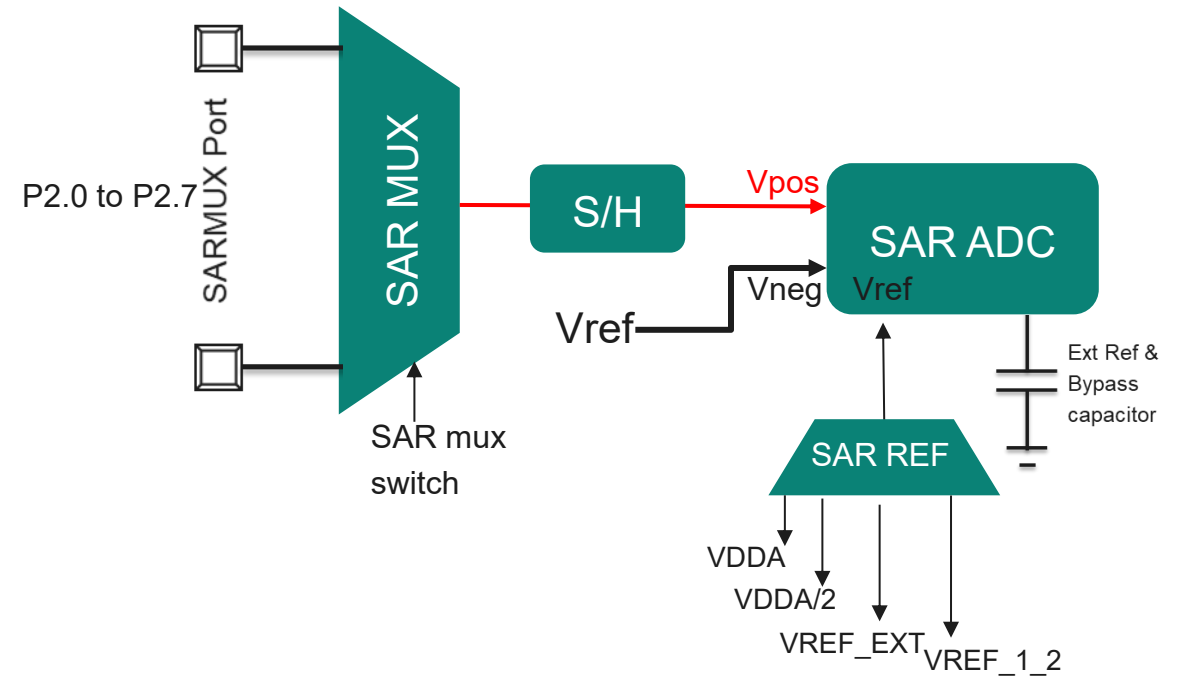
SAR ADC :

SAR ADC core is a 12-bit SAR ADC.

The maximum sample rate for this ADC is 1 Msps

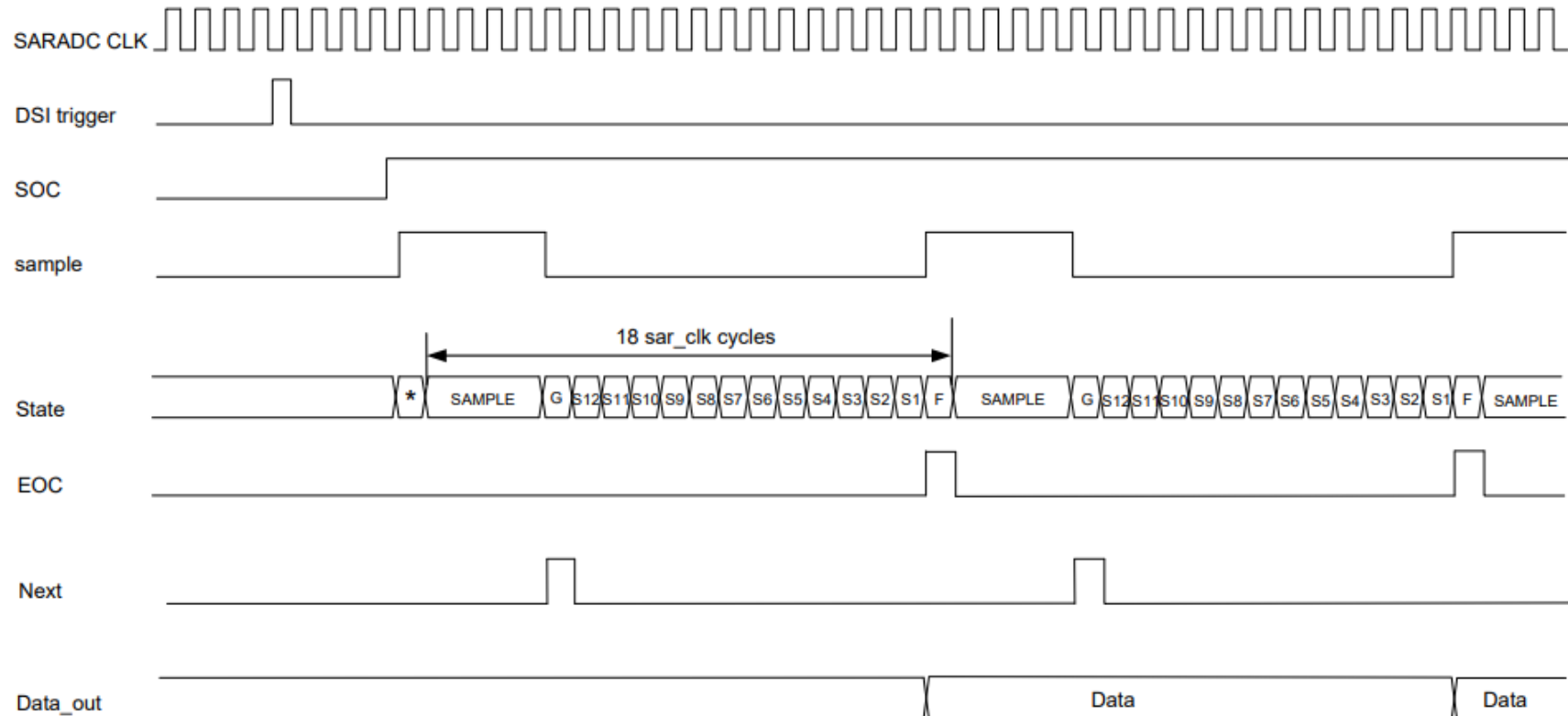
SARREF :

Provide voltage reference for SAR ADC conversion



Acquisition and Conversion time

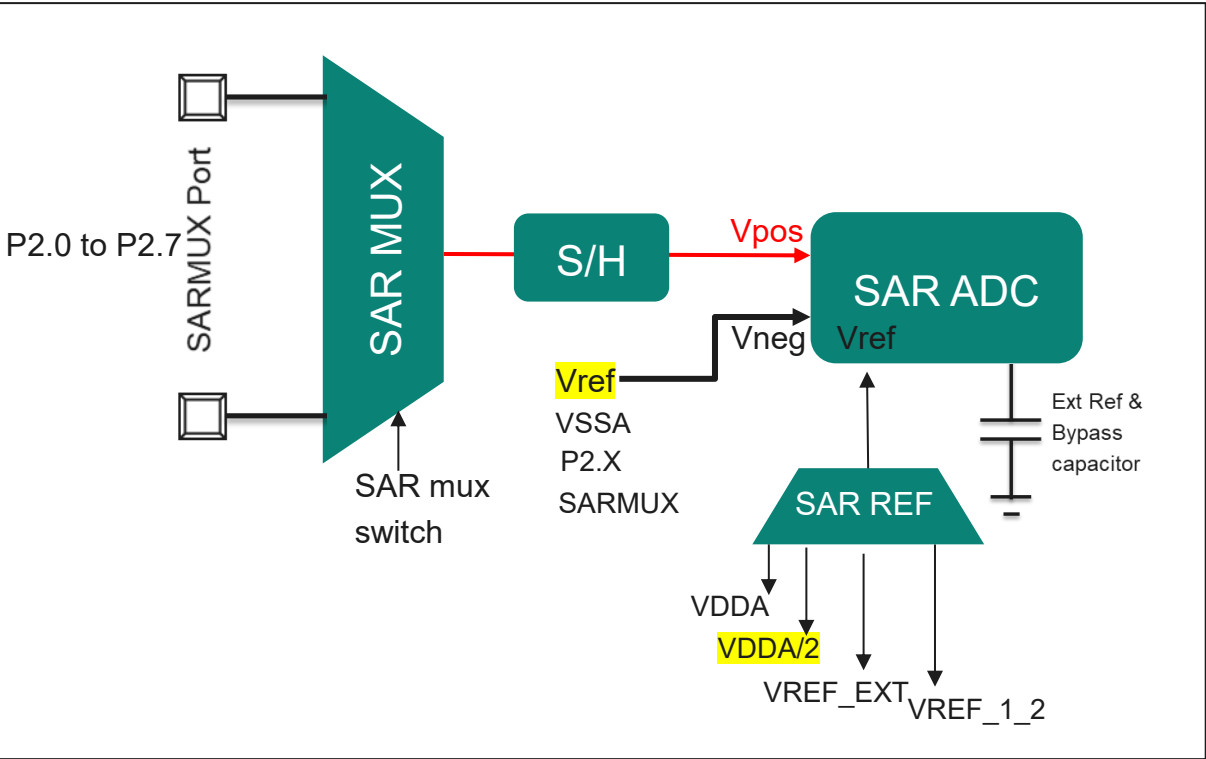
- Conversion time (sar_clk) = resolution (bit) + 2
- Total acquisition and conversion time (sar_clk) = acquisition time + resolution (bit) + 2



For 12-bit ADC with acquisition time of 10 cycles, How many cycles required for conversion ??

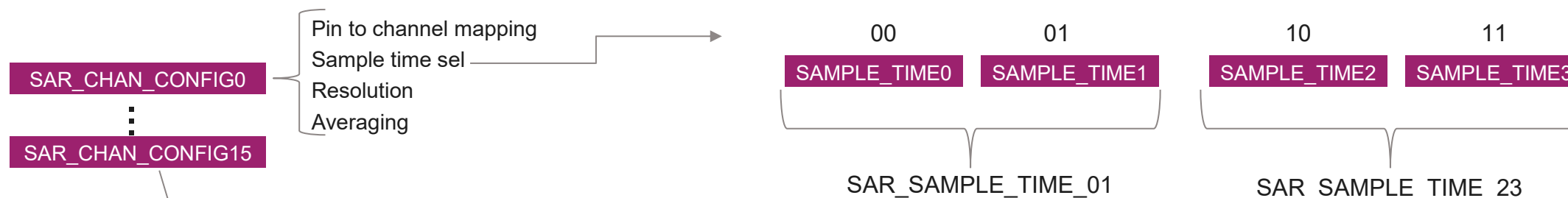
What should be the SAR Clock to get 500Ksps ??

PSOC SAR ADC PROGRAMMING MODEL



- 1 SAR_CTRL
 - Choice of Vref,
 - choice of Vneg select
 - Enable SAR ADC
- 2 SAR_MUX_SWITCH0
 - Select Port Pin for V+ of SAR
- 3 SAR_SAMPLE_CTRL
 - Choice of Left/Right Align
 - Choice of Signed/unsigned
 - Choice of Continuous/mono shot
- 4 SAR_SAMPLE_TIME01
 - ADC clock cycles for Acquisition
- 5 SAR_CHAN_CONFIG0
 - Pin to channel mapping
 - Sample time sel
 - Resolution
 - Averaging
- 6 SAR_CHAN_EN
 - Enable configured channel
- 7 SAR_START_CTRL
 - Trigger Start of Conversion
- 8 SAR_INTR
 - Wait till End of Scan (EOS)
- 9 SAR_CHAN_RESULT0
 - corresponding channel scan result

CHAN_CONFIG SAMPLE_TIME_SEL



24.1.10 SAR_CHAN_CONFIG0

Channel configuration register.

Address: 0x403A0080

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW		None	RW		None	
HW Access	None	R		None	R		None	
Name	None	PORT_ADDR [6:4]		None	PIN_ADDR [2:0]		None	

Bits	15	14	13	12	11	10	9	8
SW Access	None		RW		None	RW	RW	RW
HW Access	None		R		None	R	R	R
Name	None [15:14]		SAMPLE_TIME_SEL [13:12]		None	AVG_EN	RESOLUTION	DIFFERENTIAL_EN

24.1.3 SAR_SAMPLE_TIME01

Sample time specification ST0 and ST1

Address: 0x403A0010

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	SAMPLE_TIME0 [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:10]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	SAMPLE_TIME1 [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:26]							

Register Configuration

Common SAR Registers

Register Name	Address
SAR_CTRL	0x403A0000
SAR_SAMPLE_CTRL	0x403A0004
SAR_SAMPLE_TIME01	0x403A0010
SAR_SAMPLE_TIME23	0x403A0014

SAR_CHAN_EN	0x403A0020
SAR_START_CTRL	0x403A0024

SAR_CHAN_CONFIG0	0x403A0080
------------------	------------

SAR_CHAN_RESULT0	0x403A0180
------------------	------------

SAR_INTR	0x403A0210
----------	------------

SAR_MUX_SWITCH0	0x403A0300
-----------------	------------

Register Name	Description
SAR_CTRL	Contains ADC Configuration options like Enable bits, Reference voltage selection, Bypass selection.
SAR_SAMPLE_CTRL	Contains Sample control options like scan type(continuous/one shot), o/p data format(signed/unsigned),alignment , resolution etc
SAR_SAMPLE_TIME01	Contains sample time(aperture) specified in ADC cycles for ST0 & ST1
SAR_SAMPLE_TIME23	Contains sample time(aperture) specified in ADC cycles for ST2 & ST3
SAR_CHAN_EN	Contains one enable bit for all the SAR channels
SAR_START_CTRL	Contains Start control bits for all enabled channels (firmware trigger).
SAR_CHAN_CONFIGx	Contains selection of sample time, resolution, Diff, Port & Pin address
SAR_CHAN_RESULTx	Contains SAR conversion result of the channel. The data is copied here from the WORK field after all enabled channels in this scan have been sampled.
SAR_INTR & SAR Interrupt Registers	Interrupt management registers can be used to find out whether event is active and mask event.
SAR_MUX_SWITCH	Used to control SARMUX switch using firmware

PSOC SAR ADC PROGRAMMING SEQUENCE

- GPIO pin init for ADC sensing (select appropriate drive mode for pin (analog))
- Initialize SAR –
 - [SAR_CTRL] → Select Vref, Bypass cap, neg pin select ref, Enable SAR
 - [SAR_MUX_SWITCH0] → Connect P2.1 to Vpos of SAR ADC
 - [SAR_SAMPLE_CTRL] → Left/Right Align, Continuous/single shot, signed/unsigned
 - [SAR_SAMPLE_TIME01] → Set the Sampling time in clocks
 - [SAR_CHAN_CONFIG0] → pin to channel binding, Sample time, resolution
 - [SAR_CHAN_EN] → Enable desired channel number (chan 0)
- Start ADC conversion
 - [SAR_START_CTRL] → Trigger start of conversion
 - [SAR_INR] → check for EOS -Wait till ADC completes conversion
 - [SAR_CHAN_RESULT0] → Get the Result of ADC conversion

LDR SCH

